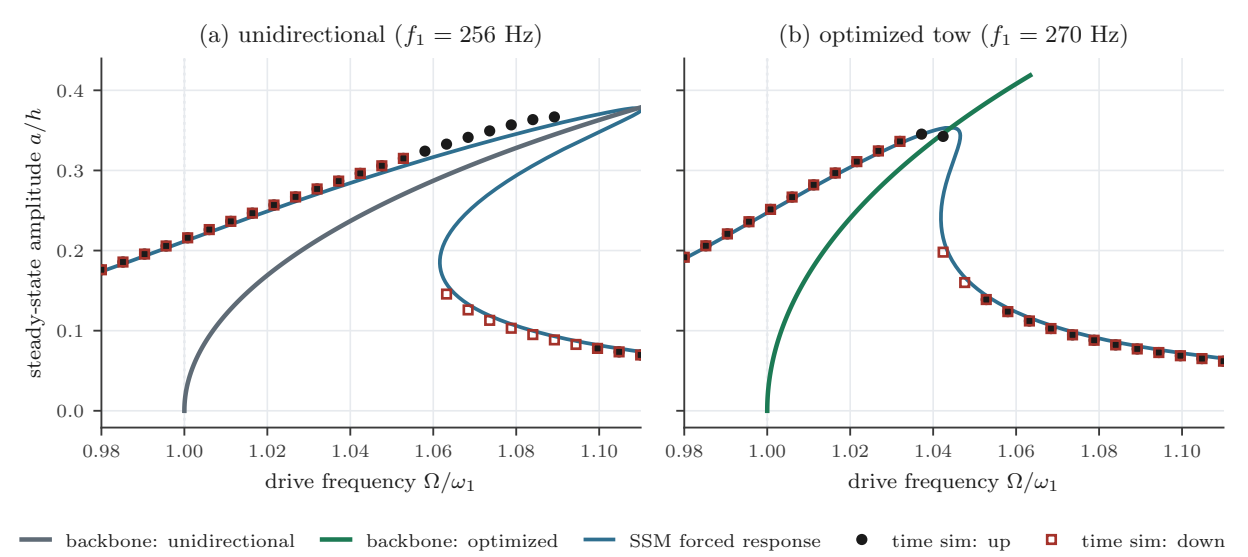

(a) unidirectional ($f_1 = 256$ Hz)
(b) optimized tow ($f_1 = 270$ Hz)
steady-state amplitude a/h
0.0
0.1
0.2
0.3
0.4
0.98
1.00
1.02
1.04
1.06
1.08
1.10
drive frequency Ω/ω_1
backbone: unidirectional
backbone: optimized
SSM forced response
time sim: up
time sim: down